



What makes African swine fever endemic in wild boars?

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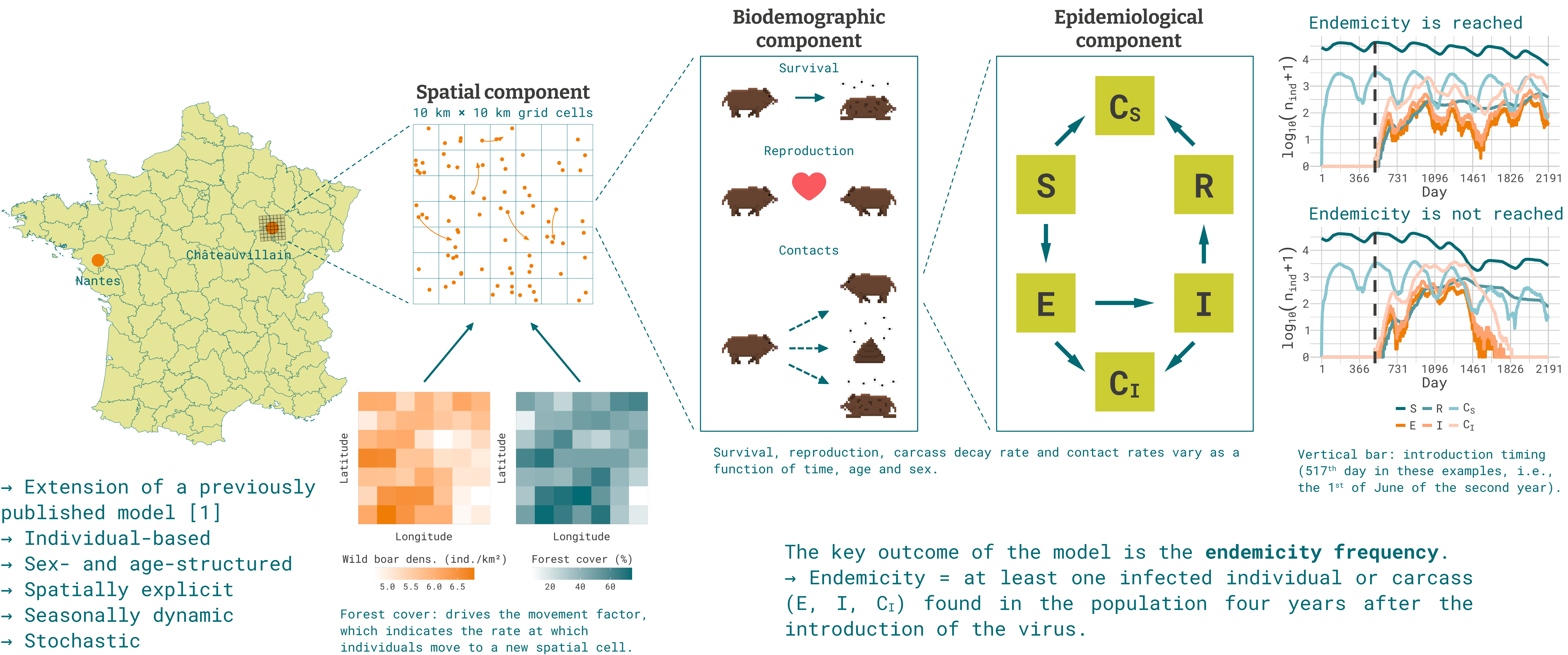
Context

- African swine fever (ASF) is a highly fatal disease infecting suids such as wild boars (*Sus scrofa*) and domestic pigs (*Sus scrofa domesticus*).
- Introduced in Europe in 2007 through Georgia, it is now spreading worldwide, rising food security and biodiversity concerns.
- Wild boars may act as a reservoir for the disease, yet **little is known on the factors leading to the establishment of the disease in wild boar populations, i.e., endemicity.**

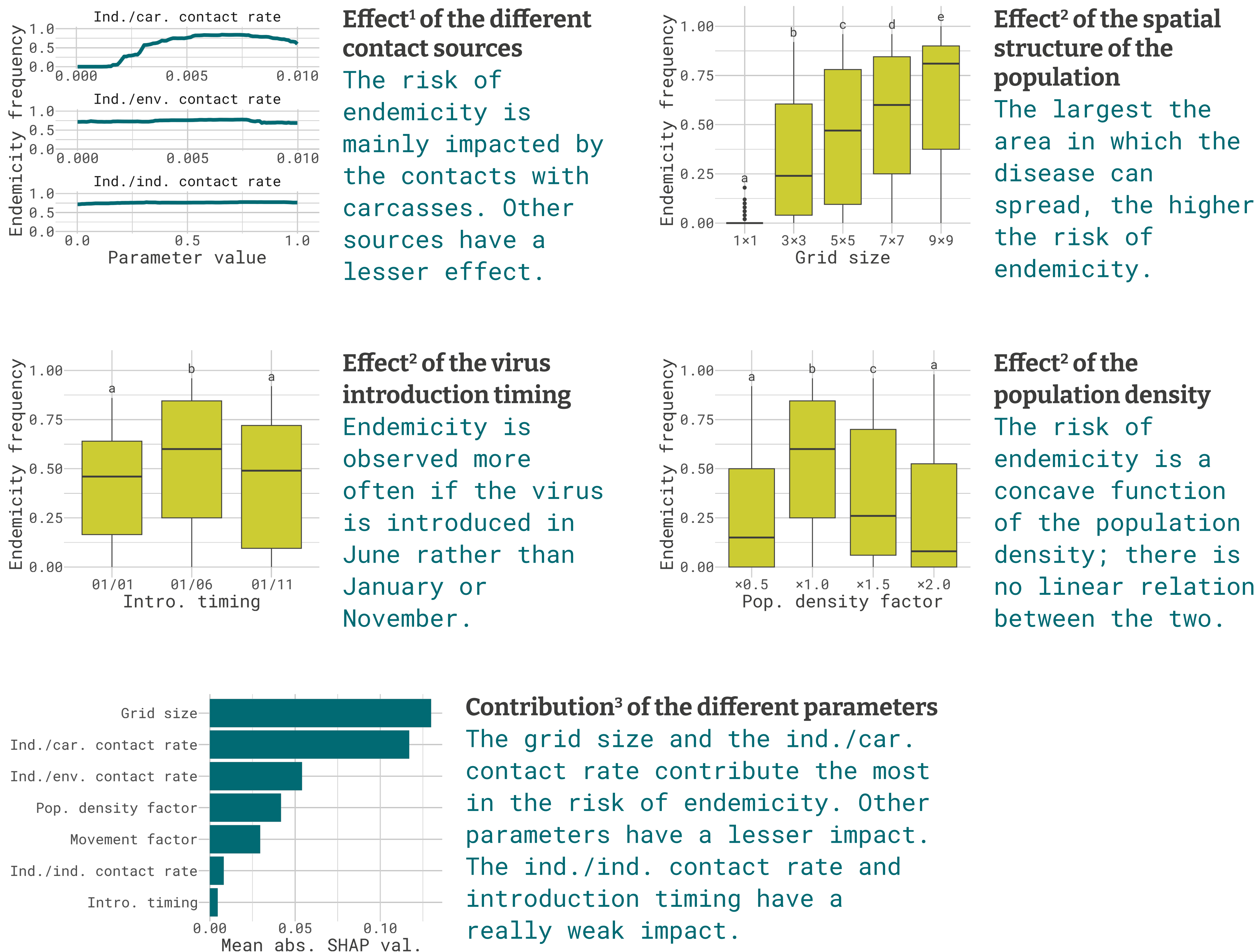


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Methods



Results & Discussion



Take-home messages

- Carcasses play a major part in driving the risk of endemicity.
- A simple increase in the population density does not lead to an increase in the risk of endemicity.
- A vast spatial structure ensures the renewal of the susceptible pool, helping the long-term establishment of the virus.

Control strategies such as **carcass removal** and **movement limitation** (fences) appear more efficient than density reduction (depopulation).

¹ Predicted endemicity frequencies obtained with a random forest regression (*randomForest* package) where the explanatory variables are the ind./ind. contact rate, the ind./car. contact rate, the ind./env. contact rate, the movement factor (not shown) and the introduction timing (not shown) and the response variable is the endemicity frequency.

² Different letters indicate significant statistical difference (i.e., $p_{adj} < 0.05$) from a pairwise chi-squared test with Bonferroni adjustment.

³ Contribution measured with SHAP values (*treeshape* package) on a global random forest regression using all the mentioned variables.

[1] R. Fay et al. (2025). "The time of infection matters: seasonal interplay between host and pathogen drives epidemic dynamics". *Oikos*, 2025(3), e10750.